

Lilygo Pager: Proximity Sensor & Relay

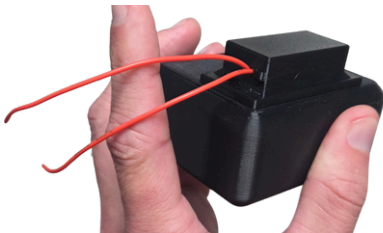
The Blackout Comms off-grid messaging & location firmware is capable of working with several add-on hardware components. The XM125 proximity/presence sensor and single relay from Sparkfun both work.



The pager can send notifications upon sensing motion, remotely control a circuit, and more.

If you don't understand what this is or how it works, DO NOT BUILD IT!

Be aware this is for experimental & educational purposes only! This is a fun/novelty feature of Blackout Comms, but can be dangerous if you don't know what you're doing. Blackout Comms may contain bugs that cause the circuit to close/open more than once, so be aware of that as well.

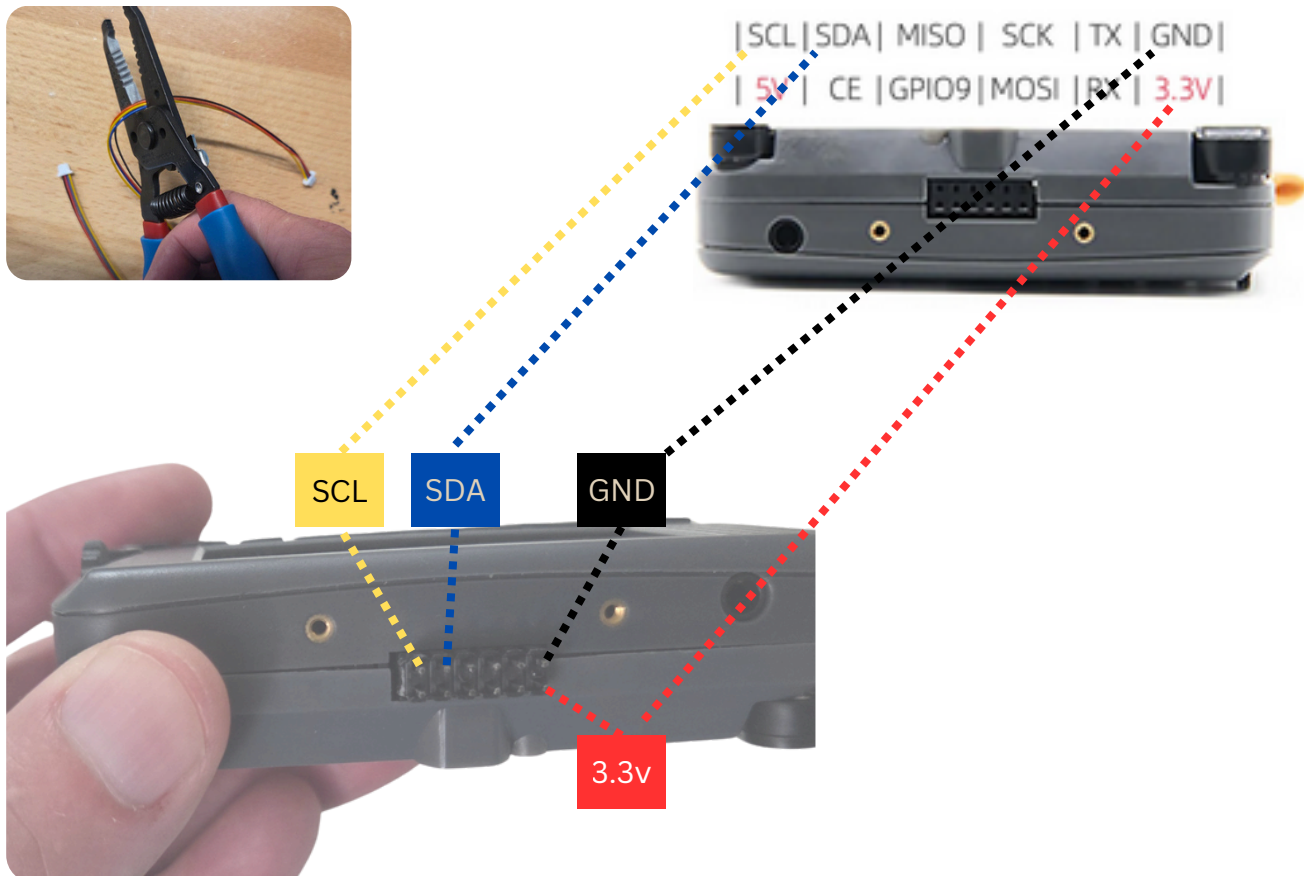
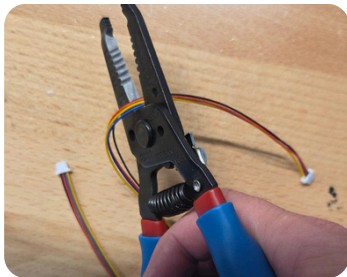


FIRST: Install Blackout Comms

Download Blackout Comms for Lilygo Pager are at:
<https://www.chatters.io/pager>

Pager Expansion Port & Header Pins

You will need to create an adapter between the pager expansion port and a standard QWIIC cable. Once this is done, the pager is ready to take Blackout Comm compatible modules.



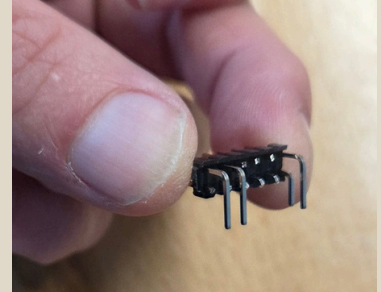
Prepare a set of header pins that match the Pager



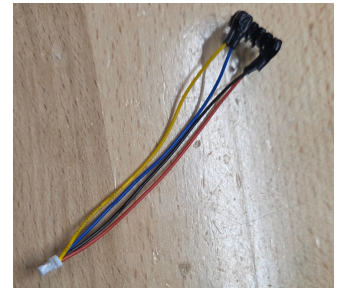
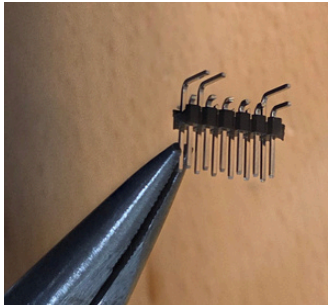
Using a 90° double-row pin header, trim it to 6x2.

Trim all the unused pins so they're not in the way.

We only need SDA, SCL, GND and 3.3v.



Solder the QWIIC Wires to the Header Pins



Use heat shrink insulator and/or liquid electrical tape to insulate the pins and bare wire, so they can't come in contact with one another.

Install the 4 Knurled Nuts

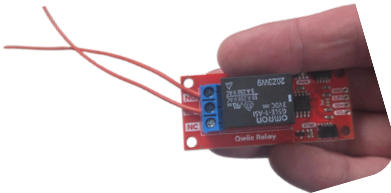


The printed backpack/housing has 6 holes for heat sinking M2 knurled nuts into.

The easiest way to sink these is with a soldering tip for melting knurled nuts into place.



Connect External Relay Wires

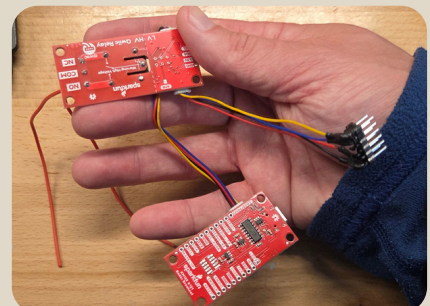


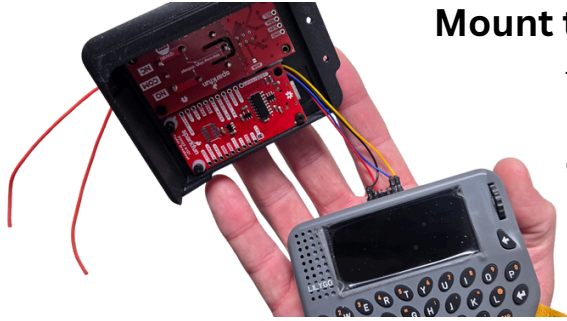
Attach two separate wires to the *Normally Open & Common* terminals of the relay.

These wires are what you'll insert into a circuit you want to be able to open/close

Connect the Header, XM125, & Relay

- Install the knurled nuts into the printed enclosure
- Connect the relay to your header pin adapter
- Connect the relay and proximity sensor using a separate QWIIC cable





Mount the Components Into Place

This is a very tight fit, and the components need to be fully inserted so they don't press into the back of the pager.

Secure the components with M2 screws (only 2-3 screws per component is necessary).

Plug the header pins into the pager, so the pins are angled down (away from the screen) as shown here.



Install the Backpack Onto the Pager

Place the backpack over the pager, careful not to pinch or trap any wires.

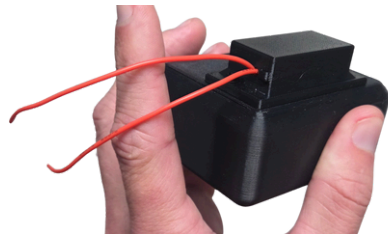
The relay & relay wires should be outside the back of the case as shown.

Use 6 M2 screws to secure.

Install the Relay Cover

Push the relay wires through the relay cover, so you can access them any time. You can zip tie them if you're not using them.

No current flows through these wires unless you supply the current via a separate circuit.



Parts List

To assemble using the instructions above, you'll need soldering equipment & supplies, as well as the components listed below:



Lilygo Pager

[Rokland, Lilygo](#)

Only compatible with SX1262 Models



SparkFun Pulsed Coherent Radar Sensor

[Sparkfun](#)



SparkFun QWIIC Single Relay

[Sparkfun](#)



QWIIC Cable

[Amazon](#)



M2 Knurled Nuts

[Amazon](#)



M2 Screws

[Amazon](#)



90 Degree 2 Row Header Pins

[Amazon](#)